Percent Yield

Calculating the percent yield of a reaction allows us to compare the theoretical yield (stoichiometric calculations) to the actual yield (lab results).

$$\%yield = \frac{actual}{theoretical} \times 100\%$$

Ex) In a laboratory experiment, 8.75g of copper is produced when 11.2 g of zinc is added to 23.9 g of copper(II) sulfate in solution. What is the percent yield of copper?

$$Zn_{(s)} + CuSO_{4(aq)} \longrightarrow ZnSO_{4(aq)} + Cu_{(s)}$$

11.2g 23.9g 8.75g

Assume $CuSO_4$ is limiting

$$m_{Zn} = 23.9g\,CuSO_4 \times \frac{1mol\,CuSO_4}{159.61g} \times \frac{1mol\,Zn}{1mol\,CuSO_4} \times \frac{65.38g}{1mol\,Zn}$$

$= 9.79g$

$\therefore CuSO_4$ is limiting

$$m_{Cu} = 23.9g\,CuSO_4 \times \frac{1mol\,CuSO_4}{159.61g} \times \frac{1mol\,Cu}{1mol\,CuSO_4} \times \frac{63.55g}{1mol\,Cu}$$

$= 9.52g$

$$\%\,yield = \frac{AY}{TY} \times 100\%$$

$$= \frac{8.75}{9.52} \times 100$$

$= 91.9\%$

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1. 3.19g / 85.6%

2. 0.413g extra
   0.749g precipitate